# Algebra 1

roger

## Contents

**1 Lecture 1: Topic** **1**
1.1 Definitions and results . . . . . 1
1.2 Examples and proofs . . . . . 1
1.3 Questions to revisit . . . . . 1
1.4 Revision summary . . . . . 1
1.5 Steinitz exchange lemma . . . . . 1

**2 Subspaces** **1**

## 1 Lecture 1: Topic

**Date:** **Key idea:**

### 1.1 Definitions and results

### 1.2 Examples and proofs

### 1.3 Questions to revisit

### 1.4 Revision summary

### 1.5 Steinitz exchange lemma

**Theorem 2.15.** *If a subspace $S$ has a spanning sequence of length $m$, then any sequence of more than $m$ elements of $S$ is linearly dependent.*

*Proof (incomplete).* Let $s_1, \ldots, s_m$ be a spanning sequence for $S$. Let $n > m$, and let $v_1, \ldots, v_n$ be any sequence of elements of $S$. We must show that $v_1, \ldots, v_n$ are linearly dependent.

*To be completed.*

**Theorem 2.17.** *Any two bases of a subspace have the same size.*

This theorem can be proven easily
about

## 2 Subspaces

Applications of linear algebra to geometry work best when the shapes involved are straight lines, plaes and their generalization s in higher dimentions.

**Definition 2.1.** (a) A set U of elements of $F^n$ is called closed under addition if for every x,y in U their sum x+y belongs to U

(b) A set U of elements of $F^n$ is called closed under scaler multiplication if for every x in

U and every number a in F, the scalar multiple ax belongs to U.

(c) A subset U of $F^n$ is called a subspace of $F^n$ if it is non-empty, closed under addition, and closed under scalar multiplication. We write $U <= F^n$ to mean that U is a subspace of $F^N$

Sometimes one subspace U is contained in another subspace V, that is every element of U also belongs to V. We write as $U <= V$ and U is a **proper** subspace if $U <= V$ and $U! = V$

Another way is

**Definition 2.2.** Let A be an $m * n$ matrix with entries in $F$.

(a) The **kernal** of A, wrriten $N(A)$, is $v \in F^n : Av = 0_m$

(b) The **column space** of A, written C(A),is $AV : V \in F^n$

**Lemma 2.3** (Kernals and column spaces are subspaces)**.** *Let $A$ be an $m * n$ matrix with entries in $F$. Then $N(A) \leq \mathbb{F}^n$ and $C(A) <= \mathbb{F}^m$.*

*Proof.* We chekc 3 subspace conditions(non-empty,closed under addition and closure under scalar multiplication).

(a) Kernal. $A0_n = 0_m$, so $0_n \in N(A)$ which is therefore non-empty. Let u,v $\in N(A)$ and let $a \in \mathbb{F}$. Then

$$\begin{aligned} A(u+v) &= Au + Av \\ &= 0_m + 0_n \\ &= 0_m \end{aligned}$$

□

section